Create a Database

```
mysql> CREATE DATABASE sqlpractice;
Query OK, 1 row affected (0.02 sec)

mysql> USE sqlpractice;
Database changed
mysql> CREATE TABLE orders (
    ->     ord_no INT PRIMARY KEY,
    ->     purch_amt DECIMAL(10,2),
    ->     ord_date DATE,
    ->     customer_id INT,
    ->     salesman_id INT
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO orders VALUES
    -> (70001,150.50,'2012-10-05',3005,5002),
    -> (70009,270.65,'2012-09-10',3001,5005),
    -> (70002,65.26,'2012-10-05',3002,5001),
    -> (70004,110.50,'2012-08-17',3009,5003),
    -> (70007,948.50,'2012-09-10',3005,5002),
    -> (70005,2400.60,'2012-07-27',3007,5001),
    -> (70008,5760.00,'2012-09-10',3002,5001),
    -> (70010,1983.43,'2012-10-10',3004,5006),
    -> (70003,2480.40,'2012-10-10',3009,5003),
    -> (70012,250.45,'2012-06-27',3008,5002),
    -> (70011,75.29,'2012-08-17',3003,5007),
    -> (70013,3045.60,'2012-04-25',3002,5001);
Query OK, 12 rows affected (0.02 sec)
Records: 12  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM orders;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70001 |    150.50 | 2012-10-05 |        3005 |        5002 |
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70004 |    110.50 | 2012-08-17 |        3009 |        5003 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
|  70010 |   1983.43 | 2012-10-10 |        3004 |        5006 |
|  70011 |     75.29 | 2012-08-17 |        3003 |        5007 |
|  70012 |    250.45 | 2012-06-27 |        3008 |        5002 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
12 rows in set (0.01 sec)
```

Q1 — WHERE

```
mysql> SELECT *
    -> FROM orders
    -> WHERE purch_amt > 2000;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
4 rows in set (0.01 sec)
```

Q2 - WHERE with DATE

```
mysql> SELECT *
    -> FROM orders
    -> WHERE ord_date='2012-09-10';
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
+--------+-----------+------------+-------------+-------------+
3 rows in set (0.01 sec)
```

Q3 — WHERE with salesman

```
mysql> SELECT *
    -> FROM orders
    -> WHERE salesman_id=5001;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
4 rows in set (0.00 sec)
```

Q4 — ORDER BY DESC

```
mysql> SELECT *
    -> FROM orders
    -> ORDER BY purch_amt DESC;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70010 |   1983.43 | 2012-10-10 |        3004 |        5006 |
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
|  70012 |    250.45 | 2012-06-27 |        3008 |        5002 |
|  70001 |    150.50 | 2012-10-05 |        3005 |        5002 |
|  70004 |    110.50 | 2012-08-17 |        3009 |        5003 |
|  70011 |     75.29 | 2012-08-17 |        3003 |        5007 |
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
12 rows in set (0.00 sec)
```

Q5 — ORDER BY

```
mysql> SELECT *
    -> FROM orders
    -> ORDER BY ord_date;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
|  70012 |    250.45 | 2012-06-27 |        3008 |        5002 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70004 |    110.50 | 2012-08-17 |        3009 |        5003 |
|  70011 |     75.29 | 2012-08-17 |        3003 |        5007 |
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
|  70001 |    150.50 | 2012-10-05 |        3005 |        5002 |
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70010 |   1983.43 | 2012-10-10 |        3004 |        5006 |
+--------+-----------+------------+-------------+-------------+
12 rows in set (0.00 sec)
```

Q6 — SUM

```
mysql> SELECT SUM(purch_amt) AS total_revenue
    -> FROM orders;
+---------------+
| total_revenue |
+---------------+
|      17541.18 |
+---------------+
1 row in set (0.01 sec)
```

Q7 — AVG

```
mysql> SELECT AVG(purch_amt) AS average_order
    -> FROM orders;
+---------------+
| average_order |
+---------------+
|   1461.765000 |
+---------------+
1 row in set (0.00 sec)
```

Q8 — MAX

```
mysql> SELECT MAX(purch_amt) AS highest_order
    -> FROM orders;
+---------------+
| highest_order |
+---------------+
|       5760.00 |
+---------------+
1 row in set (0.01 sec)
```

Q9 — MIN

```
mysql> SELECT MIN(purch_amt) AS lowest_order
    -> FROM orders;
+--------------+
| lowest_order |
+--------------+
|        65.26 |
+--------------+
1 row in set (0.00 sec)
```

Q10 — COUNT

```
mysql> SELECT COUNT(*) AS total_orders
    -> FROM orders;
+--------------+
| total_orders |
+--------------+
|           12 |
+--------------+
1 row in set (0.01 sec)
```

Q11 — GROUP BY

```
mysql> SELECT salesman_id,
    ->        SUM(purch_amt) AS total_sales
    -> FROM orders
    -> GROUP BY salesman_id;
+-------------+-------------+
| salesman_id | total_sales |
+-------------+-------------+
|        5002 |     1349.45 |
|        5001 |    11271.46 |
|        5003 |     2590.90 |
|        5005 |      270.65 |
|        5006 |     1983.43 |
|        5007 |       75.29 |
+-------------+-------------+
6 rows in set (0.01 sec)
```

Q12 — GROUP BY Customer

```
mysql> SELECT customer_id,
    ->        SUM(purch_amt) AS total_purchase
    -> FROM orders
    -> GROUP BY customer_id;
+-------------+----------------+
| customer_id | total_purchase |
+-------------+----------------+
|        3005 |        1099.00 |
|        3002 |        8870.86 |
|        3009 |        2590.90 |
|        3007 |        2400.60 |
|        3001 |         270.65 |
|        3004 |        1983.43 |
|        3003 |          75.29 |
|        3008 |         250.45 |
+-------------+----------------+
8 rows in set (0.00 sec)
```

Q13 — GROUP BY + MAX

```
mysql> SELECT customer_id,
    ->        MAX(purch_amt) AS highest_purchase
    -> FROM orders
    -> GROUP BY customer_id;
+-------------+------------------+
| customer_id | highest_purchase |
+-------------+------------------+
|        3005 |           948.50 |
|        3002 |          5760.00 |
|        3009 |          2480.40 |
|        3007 |          2400.60 |
|        3001 |           270.65 |
|        3004 |          1983.43 |
|        3003 |            75.29 |
|        3008 |           250.45 |
+-------------+------------------+
8 rows in set (0.00 sec)
```

Q14 — GROUP BY + HAVING

```
mysql> SELECT salesman_id,
    ->        SUM(purch_amt) AS total_sales
    -> FROM orders
    -> GROUP BY salesman_id
    -> HAVING SUM(purch_amt) > 3000;
+-------------+-------------+
| salesman_id | total_sales |
+-------------+-------------+
|        5001 |    11271.46 |
+-------------+-------------+
1 row in set (0.00 sec)
```

Q15 — GROUP BY + HAVING

```
mysql> SELECT customer_id,
    ->        SUM(purch_amt) AS total_purchase
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING SUM(purch_amt) > 2500;
+-------------+----------------+
| customer_id | total_purchase |
+-------------+----------------+
|        3002 |        8870.86 |
|        3009 |        2590.90 |
+-------------+----------------+
2 rows in set (0.00 sec)
```

Q16 — COUNT + HAVING

```
mysql> SELECT customer_id,
    ->        COUNT(*) AS total_orders
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING COUNT(*) > 1;
+-------------+--------------+
| customer_id | total_orders |
+-------------+--------------+
|        3005 |            2 |
|        3002 |            3 |
|        3009 |            2 |
+-------------+--------------+
3 rows in set (0.00 sec)
```

Q17 — GROUP BY + HAVING + ORDER BY

```
mysql> SELECT customer_id,
    ->        SUM(purch_amt) AS total_purchase
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING SUM(purch_amt) > 1000
    -> ORDER BY total_purchase DESC;
+-------------+----------------+
| customer_id | total_purchase |
+-------------+----------------+
|        3002 |        8870.86 |
|        3009 |        2590.90 |
|        3007 |        2400.60 |
|        3004 |        1983.43 |
|        3005 |        1099.00 |
+-------------+----------------+
5 rows in set (0.00 sec)
```

Q18 — BETWEEN + HAVING

```
mysql> SELECT customer_id,
    ->        MAX(purch_amt) AS max_purchase
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
+-------------+--------------+
| customer_id | max_purchase |
+-------------+--------------+
|        3002 |      5760.00 |
|        3009 |      2480.40 |
|        3007 |      2400.60 |
+-------------+--------------+
3 rows in set (0.00 sec)
```

Q19 — COUNT + HAVING

```
mysql> SELECT salesman_id,
    ->        COUNT(*) AS total_orders
    -> FROM orders
    -> GROUP BY salesman_id
    -> HAVING COUNT(*) >= 2;
+-------------+--------------+
| salesman_id | total_orders |
+-------------+--------------+
|        5002 |            3 |
|        5001 |            4 |
|        5003 |            2 |
+-------------+--------------+
3 rows in set (0.00 sec)
```

Q20 — MAX + GROUP BY + HAVING

```
mysql> SELECT ord_date,
    ->        MAX(purch_amt) AS highest_purchase
    -> FROM orders
    -> GROUP BY ord_date
    -> HAVING MAX(purch_amt) > 2000;
+------------+------------------+
| ord_date   | highest_purchase |
+------------+------------------+
| 2012-10-10 |          2480.40 |
| 2012-07-27 |          2400.60 |
| 2012-09-10 |          5760.00 |
| 2012-04-25 |          3045.60 |
+------------+------------------+
4 rows in set (0.00 sec)
```